Current model (ordinary-matter shell + CoE core) vs. A_F, term by term

worked sheet, 27 Sep 2026

1. Model and where each term comes from. Sedov–Taylor flow $\vec{V} = k\,\vec{x}$ (position from the CoE; Taylor 1950; Sedov 1959), decelerated by a point core M_c at the CoE and a uniform shell Ω_s between $R_{\rm in}$ and $R_{\rm out}$ (fixed in space; Earth and all sources inside it). Spherical symmetry makes the field radial, $\vec{V} = h(q)\,\vec{q}$, with $q = s/D$ (D = CoE–Earth distance). Linear perturbation of the code's integration (from $\tau_i = 0.01$, $Ht \simeq 1$):

$$\frac{h(q)}{k} \simeq 1 - \tfrac{1}{2}\Big[\underbrace{\tfrac{M_c}{M_{\rm crit}(D)}\big(\tfrac{1}{\tau_i}-1\big)q^{-3}}_{\text{core tidal term}} + \Omega_s\Big(\underbrace{\tfrac{1}{2}}_{\text{uniform: } m=0 \text{ only}} \underbrace{-\tfrac{3}{2}r_{\rm in}^2 q^{-2} + r_{\rm in}^3 q^{-3}}_{\text{hole } (r_{\rm in}=R_{\rm in}/D)}\Big)\Big], \qquad \beta \equiv \frac{h'}{h}\Big|_{q=1}, \quad \gamma \equiv \frac{h''}{h}\Big|_{q=1}.$$

Chris's A: $a_f = |\Delta\vec{V}|$/separation with sources at $R = 14.3$ Gpc, $\varepsilon \equiv R/D$, $\mu = \cos b_0 \cos\psi$, $\psi = \ell - \ell_0$ (ℓ_0 = anti-CoE direction). Exactly $a_f \propto |h(q_s)(\hat{x} + \varepsilon\hat{n}) - h(1)\hat{x}|/\varepsilon$; to first order in ε (the higher orders are the r/D terms):

$$a_f^2 \propto 1 + B\mu^2 + \varepsilon\,(P\mu + Q\mu^3), \qquad B = 2\beta + \beta^2, \quad P = \beta(3+\beta), \quad Q = (1+\beta)(\gamma+\beta) - 2\beta.$$

Take the square root, then use $\cos^2\psi = \frac{1}{2}(1+\cos 2\psi)$, $\cos^3\psi = \frac{1}{4}(3\cos\psi + \cos 3\psi)$, $\cos^4\psi = \frac{3}{8} + \frac{1}{2}\cos 2\psi + \frac{1}{8}\cos 4\psi$, $\cos^5\psi = \frac{1}{16}(10\cos\psi + 5\cos 3\psi + \cos 5\psi)$ ($b_0 = 0$ as in the code):

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi + \ldots, \quad a_2 = \frac{\frac{B}{4} - \frac{B^2}{16}}{m_0}, \quad a_1 = \frac{\varepsilon}{2m_0}\Big[P + \tfrac{3}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{16}\Big], \quad a_3 = \frac{\varepsilon}{2m_0}\Big[\tfrac{1}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{32}\Big],$$

with $m_0 = 1 + \frac{B}{4} - \frac{3B^2}{64}$. For a pure point mass ($\gamma = -4\beta$, $\beta \to 3K$) this is the earlier line-of-sight form $K[(3\mu^2-1) + \varepsilon(\frac{9}{2}\mu - \frac{15}{2}\mu^3)]$. The geometry gives only $\cos m\psi$ terms: even harmonics set by β, odd ones by εQ. *Rotation* (the sine factor, $A = a_f[1 - {\rm Amp}\sin(\psi+\varphi)]$) adds $C_1 = a_1 - {\rm Amp}\sin\varphi(1 + \frac{a_2}{2})$, $S_1 = -{\rm Amp}\cos\varphi(1 - \frac{a_2}{2})$, $C_3 = a_3 - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$, $S_3 = -\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (C_m, S_m: $\cos m\psi, \sin m\psi$ parts). **Latitude:** the code evaluates the model in the Galactic plane only (one direction per degree, no $|b|$ cut, no sector average), so the match is done the same way. A tilt b_0 would give $B \to B\cos^2 b_0$ and ε-terms $\propto \cos b_0, \cos^3 b_0$, degenerate with β and εQ; therefore $b_0 = 0$.

2. Term-by-term solution (Newtonian, series above, point-core shape $\gamma = -4\beta$; exact least-squares in brackets):

Term	A_F coefficient	Analytic expression	Solved value
2ℓ phase	$\ell_2 = 14.96°$	ℓ_0 (axis; $\beta > 0$: fast along it)	$\ell_0 = 14.96°$ (12.8°)
2ℓ amplitude	$C_2 = 0.0734$	$a_2(\beta)$	$\beta = 0.161$ (0.210)
1ℓ (rotation)	$C_1 = 0.0398$, $S_1 = 0.0343$	C_1, S_1 of Sec. 1	Amp $= -0.068$, $\varphi = 1.02$ (-0.072, 1.03)
3ℓ cos	$C_3 = -0.0292$	$a_3(\varepsilon Q) - \frac{{\rm Amp}\,a_2}{2}\sin\varphi$	$\varepsilon Q = -0.308 \Rightarrow \varepsilon = 0.348$, $D = 41$ Gpc (47)
3ℓ sin	$S_3 = 0.0130$	$-\frac{{\rm Amp}\,a_2}{2}\cos\varphi$ (check)	predicted 0.0013: **not produced**
$m = 0$	$A_0 = 0.9965$	$N[1 - \frac{1}{2}{\rm Amp}\,a_1\sin\varphi]$	$N = 0.9971$
mass	–	$\beta \simeq \frac{3K}{1-K}$, $K = \frac{1}{2}\frac{M_c}{M_{\rm crit}(D)}(\frac{1}{\tau_i} - 1)$	$M_c/M_{\rm crit}(D) \approx 0.0010$ (linear; ≈0.0017 nonlinear)

3. Input parameters implied by the term-by-term solution vs. Table 1 of universe_bestfit_thinshell.pdf. Each column keeps that fit's shell assumption. Newtonian Ω_s free: $\Omega_s = 1.012$, thin shell $R_{\rm in} = D - R$, $R_{\rm out} = D + R$. Newtonian Ω_s=0.049: $R_{\rm in} = D - 1.028\,R$, $R_{\rm out} = D + 1.475\,R$. The unknowns are solved from the Sec. 2 equations (2ℓ phase $\to \ell_0$, 2ℓ amplitude $\to \beta \to M_c/M_{\rm crit}$, $1\ell \to$ Amp, φ, 3ℓ cos $\to \varepsilon Q \to D$; C_m, S_m, m = 1–3, by least squares). The $a_m(\beta,\gamma)$ come from the code's exact $h(q)$, not the linear formula. Masses use $H_0 = 70$, $\rho_s = \Omega_s\rho_c$.

	Newtonian, Ω_s free		Newtonian, Ω_s=0.049	
Quantity	term-by-term	Table 1	term-by-term	Table 1
D/R	6.474	6.554	2.589	3.439
D	2.857×10^{24} km (92.6 Gpc)	2.892×10^{24} km (93.7 Gpc)	1.143×10^{24} km (37.0 Gpc)	1.517×10^{24} km (49.2 Gpc)
$R_{\rm in}$	78.3 Gpc	79.4 Gpc	22.3 Gpc	34.5 Gpc
$R_{\rm out}$	106.9 Gpc	108.0 Gpc	58.1 Gpc	70.3 Gpc
M_c	2.38×10^{54} kg	2.47×10^{54} kg	9.41×10^{52} kg	2.31×10^{53} kg
	1.20×10^{24} $M_\odot$	1.24×10^{24} $M_\odot$	4.73×10^{22} $M_\odot$	1.16×10^{23} $M_\odot$
$M_c/M_{\rm crit}(D)$	0.00265	0.00265	0.00164	0.00171
$M(<D)$, total	3.62×10^{56} kg	3.72×10^{56} kg	2.29×10^{54} kg	4.56×10^{54} kg
$M_{\rm shell}$, total	8.50×10^{56} kg	8.71×10^{56} kg	1.03×10^{55} kg	1.70×10^{55} kg
M within R	3.35×10^{54} kg	3.35×10^{54} kg	1.62×10^{53} kg	1.62×10^{53} kg
Ω_s	1.012[a]	1.012	0.049[a]	0.049
ℓ_0	13.2°	12.8°	12.8°	13.0°
Amp	−0.1246	−0.1240	−0.0637	−0.0622
φ	1.267 rad	1.264 rad	0.941 rad	0.929 rad
R^2 vs Migkas	0.909	0.909	0.881	0.893
R^2 vs A_F	0.987	0.987	0.970	0.977

[a] Held fixed (that column's shell assumption). The harmonic solve weights C_m, S_m equally, not the 71 points, so its R^2 can sit slightly below the point fit.

4. Model vs. generally accepted values. Model columns use the Table 1 values of the two Newtonian fits (their saved results and code; $H_0 = 70$). $R = 14.3$ Gpc is the comoving radius of the observable universe.

Quantity	Accepted value	Source	Newt. Ω_s free (0.909)	Newt. Ω_s=0.049 (0.893)
Ordinary-matter mass within R	1.51×10^{53} kg	[1][a]	3.35×10^{54} kg (22×)	1.62×10^{53} kg (1.07×)
Total matter mass within R	9.67×10^{53} kg	[1][a]	3.35×10^{54} kg[b]	1.62×10^{53} kg[b]
Ordinary-matter density ρ_b	4.20×10^{-28} kg m^{-3}	[1]	9.32×10^{-27} kg m^{-3}	4.51×10^{-28} kg m^{-3}
Ω_b	0.0493	[1]	1.012 (= Ω_s)	0.049 (= Ω_s)
Total matter Ω_m	0.3153 ± 0.0073	[1]	1.012 (= Ω_s, no dark matter)	0.049 (= Ω_s, no dark matter)
Critical density ρ_c	8.52×10^{-27} kg m^{-3} at H_0=67.36	[1]	9.20×10^{-27} kg m^{-3} at H_0=70	9.20×10^{-27} kg m^{-3} at H_0=70
H_0 [km s^{-1} Mpc^{-1}]	67.36 ± 0.54 (CMB); 73.04 ± 1.04 (local)	[1]; [2]	70 (input)	70 (input)
Earth (Sun) speed relative to the CMB	369.82 ± 0.11 km s^{-1}	[3]	$V_O = kD = 5.90\times10^6$ km s^{-1} = 19.7 c	$V_O = kD = 3.24\times10^6$ km s^{-1} = 10.8 c
ratio V_O/(CMB dipole speed)	1		15,958	8,757
Local Group speed relative to the CMB	620 ± 15 km s^{-1}	[3]	– (not modelled)	– (not modelled)
Bulk flow within $150\,h^{-1}$ Mpc	395 ± 29 km s^{-1}	[4]	5.7 km s^{-1} c	0.3 km s^{-1} c
Core mass M_c (vs. ordinary mass within R)	–		2.47×10^{54} kg (16×)	2.31×10^{53} kg (1.5×)
Mass inside D (vs. ordinary mass within R)	–		3.72×10^{56} kg (2,460×)	4.56×10^{54} kg (30×)
Age / k^{-1}	13.797 ± 0.023 Gyr	[1]	k^{-1} = 13.97 Gyr; integration age 13.83 Gyr	k^{-1} = 13.97 Gyr; integration age 13.89 Gyr

[a] Derived from [1]: $\Omega_b h^2 = 0.02237$, $\Omega_m = 0.3153$, $H_0 = 67.36$, times the volume of the $R = 14.3$ Gpc sphere. Ratios in brackets: model / accepted ordinary-matter mass within R. [b] The model has no dark matter, so total = ordinary. [c] Mean of $\Delta\vec{V}$ over a $150\,h^{-1}$ Mpc ball around Earth ($h = 0.70$), from the code's exact expansion-rate profile $h(q)$ (second order in radius/D; the first-order term averages to zero). The sine factor is not a velocity and is not included.

Sources. [1] Planck Collaboration 2020, A&A, 641, A6 (Planck 2018 VI, TT,TE,EE+lowE+lensing). [2] Riess, A. G., et al. 2022, ApJL, 934, L7 (SH0ES). [3] Planck Collaboration 2020, A&A, 641, A1 (Planck 2018 I; Local Group value after Kogut et al. 1993, ApJ, 419, 1). [4] Watkins, R., et al. 2023, MNRAS, 524, 1885 (CosmicFlows-4). Sedov–Taylor: Taylor, G. I. 1950, Proc. R. Soc. Lond. A, 201, 159; Sedov, L. I. 1959, Similarity and Dimensional Methods in Mechanics (Academic Press).